

Varun Babbar

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Summary

ML researcher (CS PhD @ Duke, Cambridge undergrad - 1st Class) with 5+ years of experience across industry and academia. First-author papers at **ICML (Oral, top 1%; Spotlight, top 2%)**, **IJCAI (Oral)**, and **JMLR**. Specialized in interpretable machine learning on tabular data (decision trees, linear models, large-scale foundation models), with an adjacent record of shipping research into production (**estimated 2M+ savings, 1 patent, 1 pending patent**).

Education

Duke University

PhD Candidate, Computer Science

Advisor: Cynthia Rudin

Durham, NC, USA

August 2023 – Current

University of Cambridge

BA + MEng in Information and Computer Engineering

Advisors: Adrian Weller (MBE), Umang Bhatt

Cambridge, United Kingdom

2018-2022

1st Class (BA), Distinction (MEng) - **Top 5% of class**

Research Experience

Duke Interpretable Machine Learning Lab

Aug 2023 – Current

Advisor: [Dr Cynthia Rudin](#)

- Designed fast algorithms to find near-optimal decision trees on noisy, high-dimensional data. **ICML 2025 Oral (top 1%)**, 3 **ICML 2026** papers, including 1 **Spotlight (top 2%)**.
- Built a toolbox for analyzing distribution shifts in tabular and vision data (**JMLR**, ~ 17% **acceptance rate**).
- Currently working on mechanistic interpretability of tabular foundation models and uncovering regimes where interpretable models perform better.

University of Cambridge: Department of Engineering

Sept 2021 – July 2022

Advisor: [Dr Adrian Weller \(OBE\)](#), [Dr Umang Bhatt](#)

- Developed human-aligned uncertainty quantification methods for deep learning models with applications to model oversight and intervention prioritization. **IJCAI Oral (top 3%)** + **NeurIPS** workshop papers.

University of Cambridge: Department of Computer Science

June 2020 – Sept 2020

Advisor: [Dr Rafal Mantiuk](#).

- Developed state-of-the-art perceptual loss functions for **image restoration tasks** (super-resolution, deblurring, denoising).
- Co-author of paper accepted at IEEE Winter Conference on Applications of Computer Vision (**WACV '22**).

Industry Experience

Applied Research Intern at Capital One

New York, United States

June 2026 - Current

- Developing foundation models for tabular and time series data.

Applied Scientist Intern at Zillow

New York, United States

May 2025 – Aug 2025

- Developed a multimodal representation learning algorithm for generating home embeddings that integrates visual, tabular, and graph features. Tested on millions of home listings across the US.

AI Scientist at JP Morgan Chase and Co

London, United Kingdom

Sept 2022 - July 2023

- Worked on large language model systems for source code understanding, with a focus on **robustness, reliability, and evaluation of model behavior**.
- Won a firmwide generative AI hackathon, published a paper at a SWE conference, developed a unit test quality assessment tool that saved the firm ~ \$2 million, contributed to [open source](#), and a pending [patent](#).

AI Scientist Intern at JP Morgan Chase and Co

London, United Kingdom

June 2021 - Sept 2021

- Developed a federated deep learning algorithm robust to distribution shifts in client datasets. Outcome: a [patent](#) and a [paper](#) accepted at SPIE Medical Imaging 2022.

Awards and Honours

YCombinator AI Startup School - only 2000 people selected worldwide

June 2025

Graduate School Fellowship @ Duke University

Aug 2023

JP Morgan Innovation Week AI Hackathon (Runner Up)

June 2023

MEng Project Prize (For a top ranked MEng Project @ Cambridge)

July 2022

Ruth Hendry Year Prize (For exemplary academic performance)

June 2022

Foundation Scholar (2x) (For 1st class performance in exams)

Oct 2021, Oct 2022

The James and Jean Bennet Prize (For exemplary academic performance)

June 2022

Hawkes' Trust Award (For exemplary academic performance)

March 2021

HackCambridge, Oxford Hackathon, HackBrunel (Winner)

Jan 2022, Nov 2018, Oct 2019

World Topper in IB School Exams (100th percentile out of ~170000 students)

May 2018

Conference, Industry, and Academic Presentations

International Conference in Machine Learning (ICML 2025), Vancouver, Canada

July 2025

New York University, Responsible Data Science Course

April 2025

Responsible AI Symposium, Duke University

Feb 2025

Duke University, Guest Lecture, MEng in AI Program

Sept 2024, Feb 2025

Duke University, Guest Lecture, Theory and Algorithms in Machine Learning

Sept 2024

JP Morgan, Innovation Week Speaker Series, London, UK

June 2023

NeurIPS 2022 Workshop on Human-in-the-Loop Learning, Virtual

Dec 2022

JP Morgan, Applied Innovation in AI Seminar, London, UK

Nov 2022

International Joint Conference on AI (IJCAI 2022), Vienna, Austria

July 2022

ICML Workshop on Distribution Free Uncertainty Quantification, Virtual

July 2022

University of Cambridge, Department of Engineering

July 2022

Service

Research Mentorship: Michael Thomas, Allen Yao, Chris Li - CS+Math Undergrads @ Duke

Jan 2025-Current

Yixiao Wang - MS in Statistics @ Duke

June 2025-Current

Program Committee: Interpretable AI Workshop @ NeurIPS

Dec 2024

Teaching Assistant: CS671: Theory and Algorithms in Machine Learning

Fall 2024

Reviewer: ECAI 2023, WANT@ICML 2024, TheWebConf2023, Journal of the ACM, AISTATS 2025, ICML 2026, NeurIPS 2026

Skills

ML/Research: PyTorch, Transformers, optimization (discrete, stochastic, constrained), statistics (Bayesian, uncertainty quantification)

Engineering: Python (NumPy, Pandas, Scikit-Learn, HuggingFace, W&B), C++, Git, Bash, LaTeX